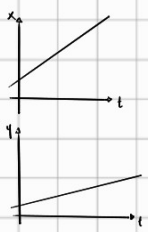
Ripasso di vettori e rendere il tutto da 1D a 2D a 3D

Moto rettilineo uniforme:

$$\begin{cases} x(t) = x_0 + v_x t \\ y(t) = y_0 + v_y t \end{cases}$$



Traiettoria

$t = \frac{x - x_0}{v_{ox}}$

$y = \frac{v_y}{v_{ox}}(x - x_0) + y_0$

Moto del proiettile

$$\begin{cases} x(t) = x_0 + v_{ox} t \\ y(t) = y_0 + v_{oy} t - \frac{1}{2} g t^2 \end{cases}$$

$v_x(t) = v_{ox} \qquad v_y(t) = v_{oy} - gt$

derivando $x(t)$ e $y(t)$